

ASSIGNMENT 2

"Predicting Banking Customer Churn: Enhancing Customer Retention Through Predictive Modelling and Analysis"

Robby Lysander Aurelio (Student ID: 32889356)

Eddy (Student ID: 33495068)

Ibra Ufauwid (Student ID: 19252994)

Master of Data Science at Monash University Indonesia



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Abstract

Customer churn refers to the rate at which customers discontinue using a company's products or services. It is a significant concern for the banking industry since it directly impacts both revenue and customer retention. This paper entitled "Predicting Banking Customer Churn: Enhancing Customer Retention Through Predictive Modelling and Analysis", investigates the creation of a predictive model for predicting client attrition for a banking institution. The project outcome of this experiment is a Customer Churn Prediction (CCP) Model.

This study aims to provide actionable insights to help the bank identify at-risk consumers by examining historical data and important criteria. In this project, the team performed some experiments of several Machine Learning (ML) models such as: Logistic Regression, Quadratic Discriminant Analysis (QDA), Decision Tree, Random Forest, and Support Vector Machine (SVM). Moreover, the experiment was performed on a dataset with a class imbalance problem, and therefore it was also conducted using techniques such as SMOTE (Synthetic Minority Over-sampling Technique) to resolve the problem. Moreover, the team also implements simple feature extraction to handle Customer Profiles in Text issues. This experiment explored predictive models that enabled banks to engage with customers more proactively, enhancing customer retention and revenue. This analysis should improve the bank's client churn management and retention strategies.

The Experiment is performed in Google Collaboration Environment, R language and an ipynb file is generated to test the models of Customer Churn Prediction (CCP). Dataset provided in csv file.

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1 Introduction

Project Description

Financial data is crucial to every organization. Data collected by financial businesses is critical for a variety of reasons. Understanding the trends, forecasting future behaviours, and making informed decisions is critical, especially in banking services. [1] In the age of big data, financial data sets are getting increasingly enormous and complicated, making it critical to efficiently use them to acquire insights and improve business outcomes. Customer engagement/customer turnover is one of the most important aspects in every banking service industry. [2]

To retain customers, bank industries must understand changing customer behaviour and expectations, especially as competition grows. There are many studies that have used data mining to forecast client churn in the banking industry. While many academics have presented numerous ways for forecasting customer churn, several Machine Learning (ML) algorithms have struggled to accurately identify churn, especially in imbalanced datasets. [3]

In this project, our team looked at a collaborative initiative aiming at creating predictive models to forecast client turnover for a bank, we named it **Customer Churn Prediction (CCP)**. The experiment uses historical data provided from the dataset, which covers variables such as credit score, tenure, balance, number of products, credit card ownership, membership participation, and salary. Furthermore, textual customer profiles combine these characteristics to present a complete picture of client behaviour. This project also faces challenges with the dataset which include class imbalance problems and managing the complexity of customer profiles in texts. Our team used techniques such as using SMOTE and implementing feature extraction to address those issues.

Project Goals

In this project, the team performed some experiments of several Machine Learning (ML) models such as: Logistic Regression, Quadratic Data Analysis (QDA), Decision Tree, Random Forest and Support Vector Machine (SVM). **Our main objective is to create an accurate binary classifier model that will properly predict if a customer would churn, allowing the bank to engage with them at the right time.**

The main goals of **Customer Churn Prediction (CCP)** projects include:

- **Develop Accurate Predictive Models, and address the challenges** (class imbalance, and textual dataset). Class imbalance occurs when one class (often the class of interest, like fraud detection, churn, or rare disease diagnosis) is significantly underrepresented compared to other classes.
- **Perform Optimization by Feature Selection and Engineering** based on experimented ML models.
- **Provide Statistical Analysis and Insights to give valuable insights and input** for bank industries specifically on: Enhancing Customer Engagement Strategies and Reduce Customer churn.

Project Team Work Allocation

Activities	Member Name and % Allocation
Business Case & Analysis	Ibra (40%), Eddy (30%), Robby (30%)
Data Wrangling + EDA	Eddy (40%), Robby (30%), Ibra (30%)
Model Development & Analysis	Robby (40%), Ibra (30%), Eddy (30%)

2 Pre-processing and features used

Data Wrangling

At the start of the project, our team found several issues with the dataset. The issues include:

- **Numerical encoding and incorrect datatype:** the values in some columns are listed using numerical encoding instead of using the real categorical meaning, e.g. 0 for “No” and 1 for “Yes”. This encoding also causes the columns to be read as numerical columns instead of categorical ones.
- **Customer’s profile:** the dataset is also provided with a short description about the customer in a textual format. Therefore, before using this column as feature, we need to pre-process it first, such as doing feature extraction.

To handle the issues listed above, our team had taken several actions:

- Decode the numerical encoding and convert them to their original representations, such as “Yes” and “No” or “Churn” and “Not Churn”.
- Convert the columns with incorrect datatype to their respective correct datatypes.
- For the customer’s profile column, our team found out that most of the important information is already provided in the other columns, except for age, gender, and country of origin. Therefore, we only need to extract the new information using Regular Expressions (REGEX), saved the extracted features on new columns, and then we can delete the original column as it is not needed anymore.

Features and Variables

Below are the features and variables included in the dataset.

Feature	Description	Type
ID	Unique identifier (excluded from modelling)	Identifier
Credit Score	Continuous variable representing the credit-worthiness of a customer	Continuous
Tenure	Number of years the customer has been with the bank	Discrete
Balance	Current account balance	Continuous
Number of Products	Number of products the customer has with the bank	Discrete
Has Credit Card	Binary indicator of whether the customer has a credit card	Binary
Is Active Member	Binary indicator of customer’s activity status	Binary
Salary	Customer’s salary information	Continuous
Age	Customer’s age (in years)	Discrete
Origin	Customer’s country of origin, includes France, Germany, and Spain	Categorical
Gender	Customer’s gender, includes Male and Female	Binary
Y	Target label describing if the customer churned or not	Binary

Exploratory Data Analysis (EDA)

After fixing the issues in the dataset, we continued to explore the data first to gain some insights about the data pattern and their relationship with each other. From this EDA process, we found some valuable insights that include:

- There is a noticeable separation between the churned and not churned customers when their age is between 40 to 65 years old. This indicates that age might be an important feature when differentiating the two classes using a machine learning model.
- Interestingly, customers who have more than 2 products in the bank will almost always churn, while customers who have exactly 2 products are almost 100% unlikely to churn. This also suggests that the number of products can be used as a good separator when doing the classification. This finding also has business implications which will be discussed later.
- The percentage of customers with a credit score more than 800 that churned is larger than customers that did not churn.

- The percentage is also larger if the customer has a balance more than 200,000 or less than 50,000 and older than 40 years old. Interestingly, for customers with zero balance, the percentage of customers that churned is less than the customers that did not churn.

More details on the EDA explanation, please refer to Appendix A and the ipynb file submitted.

3 Models

Methodologies

To model the **Customer Churn Prediction (CCP)**, 5 machine learning algorithms were tested: **Logistic Regression, QDA, Decision Tree, Random Forest, and Support Vector Machine**. The table below shows how the model works, advantages and disadvantages of each model.

Model	Description	Advantages	Disadvantages
Logistic Regression	A linear model used for binary classification that estimates the probability of an event occurring based on input features. [4]	- Simple and Easy to implement and interpret.	- Assumes a linear relationship between features and log-odds.
		- Efficient for large datasets.	- Not suitable for non-linear problems.
		- Provides probability estimates.	- Sensitive to outliers.
Quadratic Discriminant Analysis (QDA)	A classification technique that models the decision boundary as a quadratic curve by assuming features have a normal distribution with class-specific covariance matrices. [4]	- Works well with well-separated classes.	- Sensitive to multicollinearity.
		- Handles non-linear decision boundaries.	- Requires large sample sizes for accurate covariance estimates.
		- Requires fewer assumptions than LDA.	- Prone to overfitting with small datasets.
Decision Tree	A non-linear model that makes decisions by recursively splitting the data into subsets based on feature values, forming a tree-like structure. [5]	- Simple and Easy to interpret and visualize.	- Prone to overfitting, especially with deep trees.
		- Handles both categorical and numerical data.	- Sensitive to noisy data.
		- Non-parametric and doesn't require data normalization.	- Can be biased towards dominant classes.
Random Forest	An ensemble learning method that combines multiple decision trees to improve classification accuracy and reduce overfitting. [6]	- Reduces overfitting by averaging multiple trees.	- Can be computationally expensive.
		- Handles large datasets and high-dimensional data well.	- Less interpretable than single decision trees.
		- Provides feature importance scores.	- May require tuning of hyperparameters for optimal performance.
Support Vector Machine (SVM)	A classification algorithm that finds the optimal hyperplane to separate different classes in the feature space by maximizing the margin between them. [7]	- Effective in high-dimensional spaces.	- Computationally intensive for large datasets.
		- Works well with clear margin of separation.	- Requires careful tuning of hyperparameters.
		- Robust to overfitting in high-dimensional space.	- Less effective when classes overlap or in noisy datasets.

4 Experiment setups

After doing the exploratory data analysis, the team started to build the model. The goal of this study is to **predict whether the customer will churn or not based on the given predictors**. The target variable for the prediction is represented by the Y column that contains two labels, "Churn" and "Not Churn". On the other hand, there are 10 predictors (listed in earlier table) that are going to be considered. In the EDA segment, we discovered that the dataset has an imbalance class problem. Specifically, the "Not Churn" label has a significantly higher count than the "Churn" label. Therefore, it is important to consider this before beginning to construct the model. Consequently, it was essential that we implement an appropriate approach to address this imbalance in class. SMOTE (Synthetic Minority Over-sampling Technique) can be implemented to resolve the issue. By observing the pattern of the difference between a sample and its nearest neighbour, the SMOTE algorithm generates new synthetic data. The difference is then added to the sample or the value from the sample is used. [8] Here, we will also test the models with and without applying the SMOTE technique to balance the class labels, to see whether the class imbalance really does affect the model performance or not. To model the **Customer Churn Prediction (CCP)**, 5 machine learning algorithms were experimented: **Logistic Regression, QDA, Decision Tree, Random Forest, and Support Vector Machine**.

To evaluate the performance of the models created, we consider four key classification metrics: **Accuracy, Precision, Recall and F1 Score**. [9] Furthermore, given that the business case revolves around the trend of client churn, a bank would face significant financial losses if there is a substantial number of undiscovered customers that churn. Furthermore, a significant churn rate might damage the bank's reputation, perhaps resulting in a decline in new client acquisition. **Hence, we shall prioritize on Recall and Accuracy as the important measures.**

More details on the experiment coding, please refer to the ipynb file submitted.

5 Experimental results

After the model is trained with the dataset, the model is evaluated and pitted against each other. The model with the best metric will be chosen.

Without SMOTE

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8042	0.5655	0.1907	0.2852
QDA	0.8304	0.6541	0.3651	0.4686
Decision Tree	0.8594	0.7848	0.4325	0.5577
Random Forest	0.8647	0.7786	0.4744	0.5896
SVM (Radial Kernel)	0.8585	0.8376	0.3837	0.5263

With SMOTE

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.7203	0.3968	0.7023	0.5071
QDA	0.7499	0.4337	0.7232	0.5422
Decision Tree	0.8218	0.5520	0.6907	0.6136
Random Forest	0.8423	0.6098	0.6395	0.6243
SVM (Radial Kernel)	0.7908	0.4931	0.7488	0.5946

Without SMOTE

The evaluation findings for the five classification models created without SMOTE utilizing the original training data provide insights. All models have strong Accuracy ratings but weak Recall scores, none exceeding 0.5. Logistic Regression and QDA models have F1 Scores below 0.5. Only Random Forests has a higher F1 Score of 0.58, but it still falls short of 0.6.

The dataset's class imbalance causes these outcomes, which can pose various issues such as **Bias Toward Majority Class and Poor Minority Class Performance**. The Random Forest model outperformed others despite class imbalance. Each tree in Random Forest is trained with distinct data subsets via bootstrapping. This strategy accidentally exaggerates the minority class across trees, making the model more sensitive to it than other models.

With SMOTE

After performing SMOTE, from the Accuracy scores, some method performed worse than the previous model counterpart. For example, the Logistic Regression, QDA, and SVM models did not even reach 0.8 for the Accuracy

scores. However, the Recall scores also become better with all of them having the scores above 0.7, but at the cost of the Precision scores dropped down drastically. This is due to the Precision-Recall trade-off. Although considering that with the previous model, except for SVM, the Logistic Regression and QDA's scores were still worse than this. Here, we can see that by solving the class imbalance, the Precision and Recall scores could get better. In addition, the F1 Scores for those three models also go over 0.5 which signifies a better balance between the Precision and Recall scores. **Decision Tree and Random Forest both shown fair of balance between both Recall and F1 Score, which signifies a good performance in models. Between both models, decision tree offers better interpretability and visualization.**

More details on the experimental results elaboration, please refer to the ipynb file submitted.

6 Conclusion

Based on the results of this experiment, we found that a Decision Tree model is the most effective for forecasting whether a client will churn. The argument to this is because Decision Tree achieved the highest Recall score while also performing well on the other evaluation parameters. **Recall is critical for customer churn prediction because the loss generated by a customer churn is significant, therefore a bank would prefer to properly anticipate which customers will churn in order to attempt to retain them as customers.** In other words, False Negatives are more costly than False Positives, hence Recall is preferred over Precision and Accuracy. [1] **We also discovered that the most influential predictors used by the Decision Tree for making splits were age, number_of_products, and origin.** We also demonstrated how a class imbalance can significantly reduce the model's performance, showing that resolving the class imbalance is critical when developing a classification model. Therefore, implementing SMOTE was a proven solution to tackle the challenge.

In summary, based on the **Customer Churn Prediction (CCP) model** experiment we suggested some action points and recommendations that needs to be implemented by the bank industries to retain their customers:

- **Target High-Risk Segments:** The analysis found that age, the number of products owned, and credit score are all important predictors of churn. The bank should create targeted retention strategies for customers in these high-risk categories, especially those aged 40 to 60, with low credit scores, or having multiple financial products. Banks also needed to focus on customers having 3 to 4 products who mostly will churn.
- **Personalized Customer Engagement:** Utilize the insights gained from customer profile data to personalize communication and offers. Customers identified as at-risk of churning should receive personalized incentives to encourage continued engagement with the bank.
- **Proactive Retention Campaigns:** Given the higher churn rates observed among specific age groups and those with particular account balances, the bank should implement proactive retention campaigns. In addition, collect feedback too.
- **Develop a Comprehensive Customer Retention Plan:** Create a comprehensive clients retention plan that includes a mix of personalized offers, improved customer service, and targeted marketing efforts; focus on the high-risk segments.
- **Enhance Customer Experience:** Invest in technologies and strategies that enhance the overall customer experience, particularly for those identified as high risk for churn. This could include improving digital banking services, offering more personalized financial advice, and ensuring competitive product offerings.

Future Recommendation

More Robust Algorithm: To ensure the robustness of the models we propose the framework used by Chen., et al (2019). By formulating the training objective as a max-min optimization problem, the model is still able to find the split that maximizes the worst-case performance under adversarial perturbations. Another method that they proposed is to use state of the art decision tree models, such as gradient boosted tree models. Although, switching to a more sophisticated model is likely to reduce interpretability. [10]

External Data Integration: Integrate external factors such as economic indicators (e.g., inflation rates, unemployment) or regional factors that could influence customer financial behavior. These external datasets may enhance the model's ability to generalize and adapt to changing conditions.

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Appendix A – Exploratory Data Analysis (EDA)

In this project our team performs several EDA activities. **Some activities performed in EDA include: Check the Summary Dataset, Exploring the Columns, and Correlation Analysis.** This part will explore the dataset using various statistical analysis to identify interesting patterns. The analysis consists of:

- **Descriptive Statistics:** summarize the central tendencies, dispersion, and shape of the distribution using descriptive statistics. Visualisations such as boxplots, histograms, bar charts, and pie charts will also be used to help with the analysis.
- **Correlation analysis:** calculate the Pearson correlation coefficient and use visualisations to reveal the relationships between several variables. This will also help to identify whether there are any co-dependencies between the predictors.

Kindly refer to details of the EDA activities in the ipynb file submitted.

Observations from the EDA

- **Class Imbalance Data**

When we compare the number of consumers who churned and did not churn, we can see that there were far more customers who did not churn than those who did. This results in a data imbalance, which may pose issues during the modelling process. Later in the experiments, we will implement SMOTE to tackle the challenge of class imbalance data.

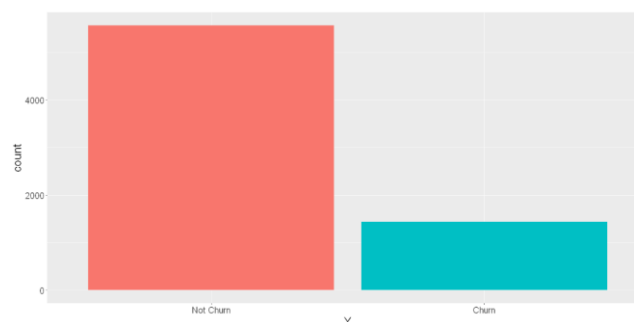


Figure 1. Chart of Customers Class Imbalance Data: Not Churn Vs Churn

- **Distribution of Data**

We observed using density chart, the age data showed a noticeable separation between the two classes (churn/not churn). The lines begin to diverge around the age of 40, suggesting a potential class separation (churn/not churn). Nevertheless, the lines reconverge at approximately 65 years of age. This significant distinction in the age column implies that it may play a significant role in the development of a classification model, as it can assist in the differentiation of the two classes.

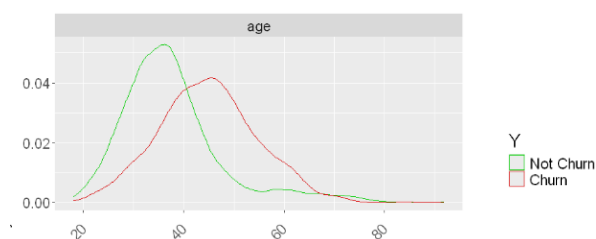


Figure 2. Density Chart of Age Data (Churn or Not Churn)

Our team also observed, based on analysis of numerical data, that it provided useful information into the tendency of customers to churn. We picked up some significant results. Active members are less likely to churn. Based on the number of products the user had, if the user had more than two products, the likelihood is that customers tended to churn. This is a significant problem that needs to be addressed by the bank.

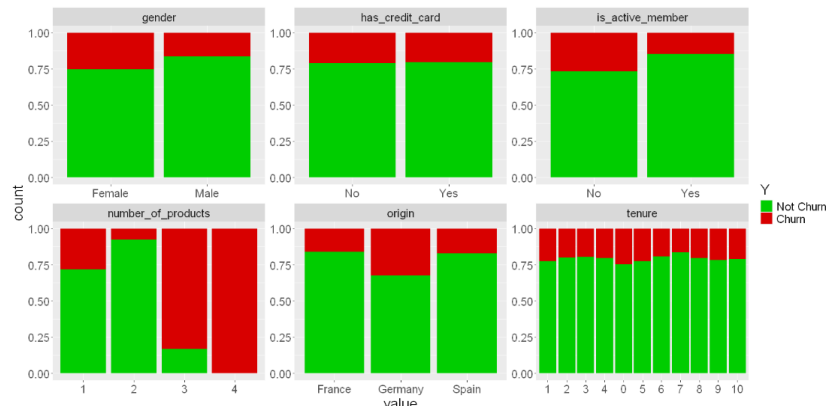


Figure 3. Bar Chart of Customers Data on Different Variables

Some scatterplots were generated in the experiments. Figure 4 showed the distribution of clients based on age and credit score. This scatterplot shows that persons with credit scores below 400 will churn. Based on the chart, many customers turnover between the ages of 40 and 60. There are some users with credit scores above 800 who also churn. See below figure.

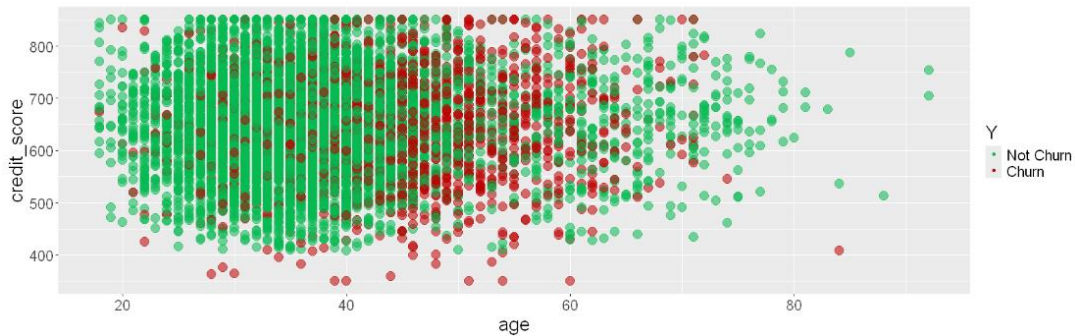


Figure 4. Scatterplot of Distribution of Clients Based on Age and Credit Score

Another similar result can be seen in the age vs. balance plot: most of the people who leave are between the ages of 40 and 60. We can also see that more customers leave when their amount exceeds 200,000. They also found an intriguing fact: users over 40 with a balance of less than 50,000 but not zero are more likely to leave. It's interesting that people who have no balance are less likely to leave. See figure 5.

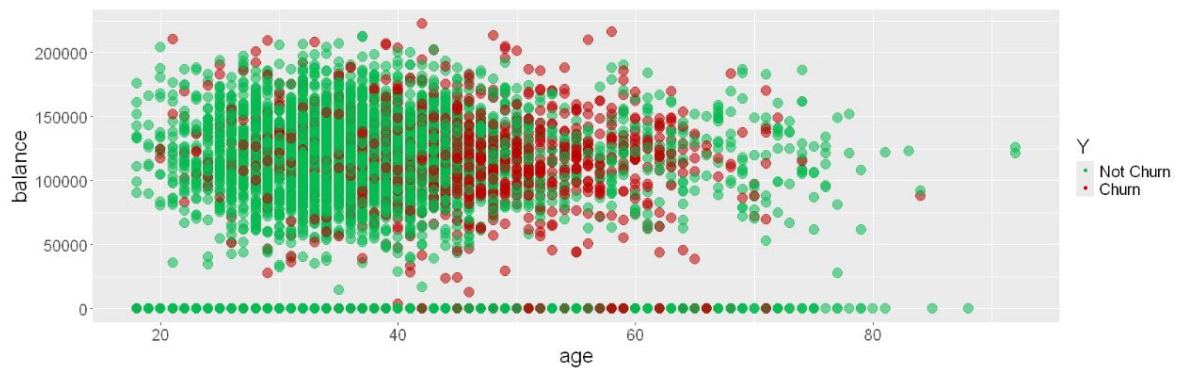


Figure 5. Scatterplot of Distribution of Clients Based on Age and Balance